Sample Complexity of Value Iteration in Reinforcement Learning with a Generative Model

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1 Introduction and Background

People make decisions every day. Sequential decision-making tasks under uncertainty are hard to solve due to partial observation, stochastic dynamics, delayed feedback, etc. A simple but powerful tool to describe such a process is *Markov Decision Process* (MDP) [Puterman, 1994]. Markov Decision Process is one of the mathematical models about the decision-making process. In particular, it assumes that the system transition is Markovian, meaning that the outcome of each decision only depends on the current state of the system. Let's formally state this tool.

1.1 Markov Decision Process

An infinite-horizon Markov Decision Process (MDP) can be describe by a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, p, r, \gamma, d_0 \rangle$. Here $\mathcal{S}$ and $\mathcal{A}$ are the state and the action space, respectively. We assume that both $\mathcal{S}$ and $\mathcal{A}$ are finite. Here p denotes the system transition probability matrix, i.e., it specifies the distribution over the next state s_{t+1} based on current state s_t and current action a_t. The quality of each decision (action) a on state s is judged by the reward function $r(s, a)$. Without loss of generality, we assume that $r(s, a) \in [0, 1]$, for all $(s, a) \in \mathcal{S} \times \mathcal{A}$ throughout. Finally, $\gamma \in [0, 1)$ is a discount factor, weighting the importance of future returns, and d_0 specifies the initial state distribution. From the view of the agent (the decision-maker), it maintains a *policy* π to select actions based on state s according to the distribution $\pi(a|s)$.

The decision process involves a series of discrete-time events: at the beginning of each episode, the system resets to state s_0 according to the initial state distribution d_0; then the agent receives the state s_0 from the system and makes an action a_0; as a result, the system transits to a new state s_1 according to transition matrix $p(s_1|s_0, a_0)$ and sends a reward $r_0 = r(s_0, a_0)$ to the agent; at this moment, the agent need to make a new action a_1 based on s_1; again, the system transits to a new state s_2 according to transition matrix $p(s_2|s_1, a_1)$ and sends a reward $r_1 = r(s_1, a_1)$ to the agent; $\cdots\cdots$ the same procedure will repeat until a terminal signal (e.g., timeout, reaching some target states, triggering some dangerous actions, etc) is released. Clearly, the terminal signal makes the length of the decision episode indeterminate, hindering an unified mathematical analysis. Luckily, we can mathematically imagine the system transits to a *absorbing state* after the terminal signal is released. Concretely, the system still runs forever on the absorbing state, but the system transits to the same absorbing state no matter which action the agent makes and the system releases a zero reward signal. In this way, the system will not stop under any conditions (hence the name infinite-horizon).

Now, we can discuss how to evaluate the quality of decision-making in an unified view. We say the agent gets a return R in terms of the cumulative rewards, i.e., $R = \sum_{t=0}^{\infty} \gamma^t r_t$. Since the system transition is probabilistic, we introduce the notation of *state value function* $V_\pi(s) = \mathbb{E}[R|s_0 = s]$ to denote the expected return it can receive if the system starts from state s, where the expectation is taken over the transition dynamics and the action selection. Similarly, we introduce the notation of *state-action value function* $Q_\pi(s, a) = \mathbb{E}[R|s_0 = s, a_0 = a]$ to denote the expected return it can receive if the system starts from state s and the agent has made the decision a. Finally, we judge the quality of policy π by its *policy value* $V_\pi \equiv \mathbb{E}_{s_0 \sim d_0}[V_\pi(s_0)]$. Importantly, we aim to find a optimal policy π^* such that it maximizes the policy value (i.e., $\pi^* = \arg\max_\pi V_\pi$).

1.2 Bellman Equation

Now, we present the most important characteristics of MDP: Bellman (Optimality) Equation:

$$\begin{aligned} V(s) &= \max_{a \in \mathcal{A}} Q(s,a), \\ Q(s,a) &= r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)}[V(s')]. \end{aligned} \tag{1.1}$$

It is proved that V^* and the associated Q^* are the unique solutions to the above Equation[Puterman, 1994]. Rearranging the above equation, we can eliminate variables and get the recursion form for V:

$$V(s) = \max_{a \in \mathcal{A}} \left[r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V(s') \right] \right]. \tag{1.2}$$

and the recursion form for Q:

$$Q(s,a) = r(s,a) + \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[\max_{a' \in \mathcal{A}} Q(s,a') \right]. \tag{1.3}$$

1.3 Bellman Operator

Based on the recursion of state value function, we can define the Bellman operator $\mathcal{T}$, which is a mapping from $\mathbb{R}^{|\mathcal{S}|}$ to $\mathbb{R}^{|\mathcal{S}|}$:

$$\mathcal{T}(V)(s) \equiv \max_{a \in \mathcal{A}} \left[r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V(s') \right] \right].$$

Similarly, we can define the Bellman operator for the state-action value function, which is a mapping from $\mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ to $\mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$:

$$\mathcal{T}(Q)(s,a) \equiv r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[\max_{a' \in \mathcal{A}} Q(s',a') \right].$$

For simplicity, we will only consider the Bellman operator for the state value function and the following properties also hold for another Bellman operator. Thus, we omit the qualifier "state" when mentioning the value function if there is no ambiguity. We will define the infinite norm as $||V||_\infty = \max_s |V(s)|$.

Lemma 1.1 (γ-contractive)**.** *For any two value function V_1 and V_2, we have*

$$\left\| \mathcal{T}(V_1) - \mathcal{T}(V_2) \right\|_\infty \leq \gamma \left\| V_1 - V_2 \right\|_\infty.$$

Proof. For any $s \in \mathcal{S}$, we have

$$\begin{aligned} |\mathcal{T}V_1(s) - \mathcal{T}V_2(s)| &= \left| \max_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_1(s') \right] \right\} - \max_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_2(s') \right] \right\} \right| \\ &\overset{(i)}{\leq} \max_{a \in \mathcal{A}} \left| \left\{ r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_1(s') \right] \right\} - \left\{ r(s,a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_2(s') \right] \right\} \right| \\ &= \gamma \max_{a \in \mathcal{A}} \left| \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_1(s') \right] - \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_2(s') \right] \right| \\ &= \gamma \max_{a \in \mathcal{A}} \left| \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[V_1(s') - V_2(s') \right] \right| \\ &\leq \gamma \max_{a \in \mathcal{A}} |V_1(s') - V_2(s')| \\ &\leq ||V_1 - V_2||_\infty, \end{aligned}$$

where step (i) uses the inequality[1] $|\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|$. The desired result follows by taking the maximum over s on the LHS. □

[1] A proof of this inequality can be founded at `https://math.stackexchange.com/questions/2041125/why-is-absolute-difference-of-max-of-functions-less-than-max-of-absolute-differe/2041131#2041131`.

Lemma 1.2 (order-reserving)**.** *For any two V_1 and V_2 satisfying that $V_1 \leq V_2$ (here $\leq$ implies V_1 is no large than V_2 elementwise), we have*

$$\mathcal{T}(V_1) \leq \mathcal{T}(V_2).$$

Proof. For each $s \in \mathcal{S}$, we have

$$\begin{aligned}
\mathcal{T}(V_1)(s) - \mathcal{T}(V_2)(s) &= \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V_1(s')\right] \right\} - \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V_2(s')\right] \right\} \\
&\overset{(i)}{\leq} \max_{a\in\mathcal{A}} \left\{ \left[r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V_1(s')\right] \right] - \left[r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V_2(s')\right] \right] \right\} \\
&= \gamma \max_{a\in\mathcal{A}} \left\{ \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V_1(s') - V_2(s')\right] \right\} \\
&\overset{(ii)}{\leq} 0,
\end{aligned}$$

where step (i) uses the inequality that $\max_x f(x) - \max_x g(x) \leq \max_x \{f(x) - g(x)\}$, and step (ii) follows the assumption that $V_1(s') \leq V_2(s')$ for every s'. □

Finally, it is clear that V^* is the fixed point of $\mathcal{T}$, that is,

$$V^* = \mathcal{T}(V^*).$$

1.4 Value Iteration

In this section, we present the fundamental algorithm called *Value Iteration* (VI) that we will analyze throughout. In the high-level, this method iteratively applies Bellman operator to update. Because Bellman operator is γ-contractive, this algorithm will convergence to the fixed point V^* finally. The procedure of this algorithm is outlined in Algorithm 1.

Algorithm 1 Value Iteration

Input: initial value V_0 and iteration number L.

```
Initialize an auxiliary variable Z ∈ R^{|S|}.
for ℓ = 1, 2, ··· , L do
    for each state s ∈ S do
        Z(s) = max_{a∈A} [r(s,a) + γ E_{s'∼p(·|s,a)} [V_{ℓ-1}(s')]].
    end for
    Set V_ℓ = Z.
end for
```

Output: V_L.

Theorem 1.3 (Linear Convergence of Value Iteration)**.** *With the parameter*

$$L = \left\lceil \frac{1}{1-\gamma} \log \frac{1}{\epsilon} \right\rceil,$$

Value Iteration (see Algorithm 1) can find a sub-optimal value function V_L such that $||V_L - V^*||_\infty \leq \epsilon$ from any initial solution V_0, where $\epsilon \in (0, \frac{1}{1-\gamma}]$ is the error tolerance.

Proof. We can see that Algorithm 1 applies $\mathcal{T}$ each iteration. Thus, we have that $V_\ell = \mathcal{T}V_{\ell-1}$.

$$\|V_\ell - V^*\|_\infty = \|\mathcal{T}V_{\ell-1} - \mathcal{T}V^*\|_\infty \leq \gamma \|V_{\ell-1} - V^*\|_\infty$$

where the inequality uses the γ-contractive of $\mathcal{T}$ (see Lemma 1.1). Applying this inequality repeatedly, we have

$$\|V_L - V^*\|_\infty \leq \gamma^L \|V_0 - V^*\|_\infty \leq \frac{\gamma^L}{1-\gamma},$$

where we use the property that $\|V_0 - V^*\|_\infty \leq \frac{1}{1-\gamma}$.

$$\frac{\gamma^L}{1-\gamma} \leq \epsilon \Longrightarrow L \geq \frac{1}{\log \frac{1}{\gamma}} \log \frac{1}{\epsilon}.$$

Finally, we note that $\log x \leq x - 1$ for any $x > 0$. $\square$

2 Sampling-based Value Iteration

In this section, we consider a new setting, where we do not know the system transition p in advance. However, we may sample each transition tuple (s, a, r, s') by taking action a at state s and getting reward r and the next state s'. This setting is closely related to reinforcement learning. But it differs from RL with the feature that we can directly go (reset) to any state s without the planning behaviors from the initial state s_0. This setting is called **Generative Model**, which is formally defined below.

Definition 2.1 (Generative Model)**.** *A generative model $\mathcal{M}$ is an MDP with the additional opportunity that we can directly reset to any state s_t, upon which we can take any action a_t to get reward $r(s_t, a_t)$ and the next state according to $p(\cdot|s_t, a_t)$.*

An example of a generative model is the game simulator, where we can reset the simulator to any frame by loading the RAM.

Algorithms based on the setting of the generative model are called **sampling-based methods** because we can only collect samples to get information about system transition. Here algorithms are often classified into two classes: model-free methods and model-based methods. Model-free methods don't construct a (virtual) MDP explicitly. Instead, they replace the population-based Bellman operator with the empirical one. On the other hand, model-based approaches explicitly construct a (virtual) MDP and subsequently run algorithms like value iteration on the covered MDP since the covered population Bellman operator is available. Readers may want to know the advantages and disadvantages of the two methods. But before discussing this, let's consider the complexity of algorithms in this setting.

Similarly, we still hope to find an ϵ-optimal state value function V or an ϵ-optimal policy. However, costs to achieve this in the generative model setting are closely related to the calls of the generative model oracle. The number of calls is often called the number of samples in the literature of machine learning. With more calls, we can get more information about the system and quickly find an ϵ-optimal solution. However, it is usually expensive to call such an oracle. Thus, we expect good algorithms don't waste samples and enjoy a good sample complexity. In addition, we notice that randomness is injected to algorithms due to the return of the oracle is stochastic. Therefore, we also hope our algorithm can success with a high probability. A formal definition is given below.

Definition 2.2 (Sample Complexity on Generative Model)**.**

2.1 Model-free Value Iteration

In this section, we consider a simple model-free algorithm called *Phased Value Iteration*. Compared with standard value iteration, it replaces the population-based Bellman operator with the empirical one. The procedure of this algorithm is outlined in Algorithm 1. At a high level, each iteration of this algorithm samples m next states to approximate the population-based Bellman operator used in the standard algorithm of *Value Iteration* and uses these samples to update the value function.

To analyze the algorithm of *Phased Value Iteration*, we recall that the iterator V_ℓ used in the algorithm of *Value Iteration* geometrically converges to the optimal V^* . Therefore, the iterator $\hat{V}_\ell$ can also enjoy a similar convergence result if $\hat{V}_\ell$ is close to V_ℓ , which can be achieved by choosing a large n. Actually, we only need to ensure the number of samples n is large enough to guarantee $\sum_{i=1}^{n} \frac{1}{n}\hat{V}_\ell(s'_i)$ is close to $\mathbb{E}_{s'\sim p(\cdot|s,a)}[V_\ell(s')]$. In addition to the choice of n, the iteration number T can be determined to be $\lceil \frac{\log((1-\gamma)/\epsilon)}{\log(1/\gamma)} \rceil$, the one we used in value iteration. Finally, V_0 can be initialized to the zero vector for simplicity.

Algorithm 2 Phased Value Iteration

Input: initial value V_0, iteration number L, and sample size n.

Initialize $\hat{V}_0 = V_0$ and a policy $\hat{\pi}_0 \in \mathbb{R}^{|\mathcal{S}|}$.
Initialize an auxillary variable $Z \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$.
for $\ell = 1, 2, \cdots, L$ **do**
  for each state $s \in \mathcal{S}$ **do**
    for each any $a \in \mathcal{A}$ **do**
      Sample n next states $\{s_i'\}_{i=1}^n$ by calling $\mathcal{M}$.
      Set $Z(s,a) = r(s,a) + \gamma \sum_{i=1}^n \frac{1}{n} \hat{V}_{\ell-1}(s_i')$.
    end for
    Set $\hat{V}_\ell(s) = \max_{a \in \mathcal{A}} Z(s,a)$ and $\hat{\pi}_\ell(s) = \arg\max_{a \in \mathcal{A}} Z(s,a)$.
  end for
end for

Output: $(\hat{V}_L, \hat{\pi}_L)$.

Theorem 2.3 (Sample Complexity of Phased Value Iteration)**.** *Given a generative model $\mathcal{M}$, with the parameters:*

$$T = \left[\frac{1}{1-\gamma} \log \frac{2}{(1-\gamma)\epsilon}\right], \quad n = \frac{4}{\epsilon^2 (1-\gamma)^4} \log\left(\frac{2 \times |\mathcal{S}| \times |\mathcal{A}| \times T}{\delta}\right), \quad V_0 = 0$$

Phased Value Iteration (see Algorithm 2) ensures that $||\hat{V}_T - V^*||_\infty \leq \epsilon$ with probability at least $1 - \delta$, and the total number of samples used is

$$\mathcal{O}\left(\frac{|\mathcal{S}| \times |\mathcal{A}|}{\epsilon^2 (1-\gamma)^5} \log\left(\frac{|\mathcal{S}| \times |\mathcal{A}|}{\delta}\right)\right).$$

Proof. The proof mainly refers to [Kearns and Singh, 1999, Kakade, 2003] and involves two steps. In the first step, we analyze the error bound $\|\hat{V}_T - V^*\|_\infty$ with the assumption that the sampling-based Bellman update is accurate in the sense that $|\sum_{i=1}^n \frac{1}{n} \hat{V}(s_i') - \mathbb{E}_{s' \sim p(\cdot|s,a)}[V(s')]| \leq \epsilon_n$. In the second step, we use concentration inequalities to properly choose the main parameter n to ensure the error is small with high probability.

Suppose that for each state-action pair (s,a) and each iteration ℓ, we have

$$\left|\sum_{i=1}^n \frac{1}{n} \hat{V}_{\ell-1}(s_i') - \mathbb{E}_{s' \sim p(\cdot|s,a)}\left[\hat{V}_{\ell-1}(s')\right]\right| \leq \epsilon_n. \tag{2.1}$$

In this way, we can show that $\hat{V}_\ell$ is close to V_ℓ. To simplify notation, we will denote $\sum_{i=1}^n \frac{1}{n} \hat{V}_{\ell-1}(s_i')$ by

$\mathbb{E}_{s'\sim\hat{p}(s,a)}[\hat{V}_{\ell-1}(s')]$, where $\hat{p}(s'|s,a) = \frac{\#\text{ times}(s,a)\mapsto s'}{n}$. Then for each $s \in \mathcal{S}$, we have

$$\begin{aligned}
&\left|V_\ell(s) - \hat{V}_\ell(s)\right| \\
&= \left|\max_{a\in\mathcal{A}} \left\{r(s,a) + \gamma\mathbb{E}_{s'\sim p(\cdot|s,a)}\left[V_{\ell-1}\left(s'\right)\right]\right\} - \max_{a\in\mathcal{A}} \left\{r(s,a) + \gamma\mathbb{E}_{s'\sim\hat{p}(\cdot|s,a)}\left[\hat{V}_{\ell-1}\left(s'\right)\right]\right\}\right| \\
&\overset{(i)}{\leq} \max_{a\in\mathcal{A}} \left|\left\{r(s,a) + \gamma\mathbb{E}_{s'\sim p(\cdot|s,a)}\left[V_{\ell-1}\left(s'\right)\right]\right\} - \left\{r(s,a) + \gamma\mathbb{E}_{s'\sim\hat{p}(\cdot|s,a)}\left[\hat{V}_{\ell-1}\left(s'\right)\right]\right\}\right| \\
&= \gamma \max_{a\in\mathcal{A}} \left|\mathbb{E}_{s'\sim p(\cdot|s,a)}\left[V_{\ell-1}\left(s'\right)\right] - \mathbb{E}_{s'\sim\hat{p}(\cdot|s,a)}\left[\hat{V}_{\ell-1}\left(s'\right)\right]\right| \\
&\overset{(ii)}{\leq} \gamma \max_{a\in\mathcal{A}} \left|\mathbb{E}_{s'\sim p(\cdot|s,a)}\left[V_{\ell-1}\left(s'\right)\right] - \mathbb{E}_{s'\sim p(\cdot|s,a)}\left[\hat{V}_{\ell-1}\left(s'\right)\right]\right| + \gamma\epsilon_n \\
&\leq \gamma \max_{s'\in\mathcal{S}} \left|V_{\ell-1}\left(s'\right) - \hat{V}_{\ell-1}\left(s'\right)\right| + \gamma\epsilon_n,
\end{aligned} \tag{2.2}$$

where step (i) again uses the inequality $|\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|$, and step (ii) uses the triangle inequality. By repeatedly applying the above inequality, we have that for each iteration ℓ

$$\|V_\ell - \hat{V}_\ell\|_\infty \leq \gamma^\ell \|V_0 - \hat{V}_0\|_\infty + \sum_{i=1}^{\ell} \gamma^i \epsilon_n \leq \frac{\gamma}{1-\gamma}\epsilon_n,$$

By triangle inequality, we have that for each iteration ℓ,

$$\left\|V^* - \hat{V}_\ell\right\|_\infty \leq \|V^* - V_\ell\|_\infty + \left\|V_\ell - \hat{V}_\ell\right\|_\infty \leq \gamma^\ell \|V_0 - V^*\|_\infty + \frac{\gamma}{1-\gamma}\epsilon_n \leq \frac{\gamma^\ell + \gamma\epsilon_n}{1-\gamma}.$$

Now suppose we can properly choose n such that $\gamma\epsilon_n \leq \gamma^L$. In this way, we show that

$$\left\|V^* - \hat{V}_L\right\| \leq \frac{2\gamma^L}{1-\gamma}.$$

By setting L such that the RHS is less than the desired error ϵ, we obtain that

$$L \geq \frac{1}{1-\gamma}\log\frac{2}{(1-\gamma)\epsilon}.$$

Note that $\hat{V}_\ell$ is bounded in the interval $[0, \frac{1}{1-\gamma}]$ for $\ell = 1, 2\cdots, L$. Therefore, we can use Hoeffding's inequality to obtain that

$$\Pr\left(\left|\sum_{i=1}^{n} \frac{1}{n}\hat{V}_{\ell-1}\left(s_i'\right) - \mathbb{E}_{s'\sim p(\cdot|s,a)}\left[\hat{V}_{\ell-1}\left(s'\right)\right]\right| \geq \epsilon_n\right) \leq 2\exp\left(-2n(1-\gamma)^2\epsilon_n^2\right).$$

By setting RHS equal to the fail probability δ', we have that when $n = \frac{1}{(1-\gamma^2)\epsilon_n^2}\log\frac{2}{\delta'}$, with probability at least $1-\delta'$,

$$\left|\sum_{i=1}^{n} \frac{1}{n}\hat{V}_\ell\left(s_i'\right) - \mathbb{E}_{s'\sim p(\cdot|s,a)}\left[\hat{V}\left(s'\right)\right]\right| \leq \epsilon_n.$$

Now we set $\delta' = \frac{\delta}{|\mathcal{S}|\times|\mathcal{A}|\times L}$ to ensure that this relation holds over all state-action pairs and L iterations with at least probability $1-\delta$. Finally, note that $\epsilon_n \leq \gamma^L \leq \frac{\epsilon}{2(1-\gamma)}$, we know the following parameters guarantee $\hat{V}_L$ is ϵ-suboptimal with probability at least $1-\delta$,

$$L = \left[\frac{1}{1-\gamma}\log\frac{2}{(1-\gamma)\epsilon}\right], \quad n = \frac{4}{\epsilon^2(1-\gamma)^4}\log\left(\frac{2\times|\mathcal{S}|\times|\mathcal{A}|\times L}{\delta}\right).$$

At last, the total samples used by this algorithm can be computed using the fact that each iteration requires $|\mathcal{S}|\times|\mathcal{A}|\times n$ samples and the total iteration number is L. □

Remark. We can see that *Phased Value Iteration* requires the sampling-based Bellman update is accurate for all iterations. And for each iteration, it collects new samples to estimate the expected value function. Intuitively, the past data is not reused and this algorithm is sample-inefficient.

2.2 Model-based Value Iteration

In this section, we consider a new approach, which first constructs a (virtual) MDP and subsequently runs the standard algorithm *Value Iteration* on this recovered MDP. Compared with the model-free approach, this algorithm does not repeatedly interact with the generative model. This approach called *Model-based Value Iteration* is outlined in Algorithm 3. Note that we are assumed that the reward function is known for simplicity[2].

Algorithm 3 Model-based Value Iteration

Input: n.

1: Collect n next states for each state-action pair by calling the generative model $\mathcal{M}$.

2: Construct a virtual MDP with $\hat{P}$:

$$\hat{P}(s'|s,a) = \frac{\#\text{ times } (s,a) \mapsto s'}{n}.$$

3: $\hat{V}^* \leftarrow$ Run *Value Iteration* (see Algorithm 1) on the virtual MDP.

Output: $\hat{V}^*$.

For simplicity, we assume that we can get the optimal solution to the recovered MDP. If this step is not perfect, the following analysis can be extended in a similar way (i.e., we can only get a sub-optimal solution and this suboptimal gap can be reduced by increasing the iteration numbers on the recovered MDP). To analyze the *Model-based Value Iteration* algorithm, we note that we are still interested in the gap $\|V^* - \hat{V}^*\|_\infty$. However, we can only ensure that $|\mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')] - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,a)}[V(s')]|$ is small for an arbitrary value function V by choosing a large n. Therefore, the proof of the following theorem relies on a triangle inequality to build the desired error bound.

2.2.1 Coarse Analysis

Theorem 2.4 (Sample Complexity of Model-based Value Iteration (Coarse Analysis))**.** *Given a generative model $\mathcal{M}$, with the parameter:*

$$n = \left[\frac{1}{\epsilon^2(1-\gamma)^4}\log\left(\frac{2\times|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right],$$

Model-based Value Iteration (see Algorithm 3) can find a sub-optimal value function $\hat{V}^*$ such that $||\hat{V}^* - V^*||_\infty \le \epsilon$ from any initial solution V_0, where $\epsilon \in (0, \frac{1}{1-\gamma}]$ is the error tolerance with probability at least $1-\delta$, and the number of total samples used is

$$\mathcal{O}\left(\frac{|\mathcal{S}|\times|\mathcal{A}|}{\epsilon^2(1-\gamma)^4}\log\left(\frac{|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right).$$

[2]If the reward function is unknown, we can also recover the reward function from collected samples, and it is easy to show that the total sample complexity is still dominated by the one used to recover the transition.

Proof. The proof mainly refers to [Kearns and Singh, 1999, Kakade, 2003] and involves two steps. In the first step, we assume that recovered MDP is accurate in the sense that $|\mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')]-\mathbb{E}_{s'\sim \hat{p}(\cdot|s,a)}[V(s')]| \leq \epsilon_n$. And, similar to the proof of *Phased Value Iteration* (see Algorithm 2), we use a triangle inequality to derive the error bound. Finally, we properly choose the parameter n to ensure that the error is small with high probability.

Let's start with the assumption that for each state-action pair (s, a) we have

$$\left|\mathbb{E}_{s'\sim p(\cdot|s,a)}\left[\hat{V}^*\left(s'\right)\right] - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,a)}\left[\hat{V}^*\left(s'\right)\right]\right| \leq \epsilon_n. \tag{2.3}$$

Following the same lines in the proof of *Phased Value Iteration*, we have that for each $s \in \mathcal{S}$,

$$\left|V^*(s) - \hat{V}^*\left(s'\right)\right| \leq \gamma \max_{s'\in\mathcal{S}} \left|V^*(s) - \hat{V}^*\left(s'\right)\right| + \gamma\epsilon_n. \tag{2.4}$$

Thus, we have that

$$\left\|V^* - \hat{V}^*\right\|_\infty \leq \frac{\gamma\epsilon_n}{1-\gamma}.$$

To ensure the optimality gap is less than ϵ, we need to ensure that

$$\epsilon_n \leq \frac{(1-\gamma)\epsilon}{\gamma}.$$

Note that $\hat{V}^*$ is bounded in the range $[0, \frac{1}{1-\gamma}]$. With Hoeffding's inequality, we can choose the parameter

$$n = \frac{1}{\epsilon_n^2(1-\gamma)^2}\log\left(\frac{2\times|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right),$$

to ensure that the above assumption (see Equation (2.3)) with probability at least $1-\delta$. Finally, we conclude that if we are given

$$n = \left[\frac{1}{\epsilon^2(1-\gamma)^4}\log\left(\frac{2\times|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right]$$

calls to the generative oracle, the output of Model-based Value Iteration is ϵ-optimal with probability at least $1-\delta$. □

Remark. In fact, the sample complexity of *Model-based Value Iteration* is better than the one of *Phased Value Iteration*. Technically, one can understand this result as *Model-based Value Iteration* only requires the sampling-based Bellman update is accurate for $\hat{V}^*$, while *Phased Value Iteration* requires the sampling-based Bellman update is accurate for all possible iterators $\hat{V}_t$.

2.2.2 Refined Analysis

In this part[3], we present a tighter bound for Algorithm 3. The key technical ingredient is to use Bernstein's inequality [Wainwright, 2019] rather than Hoeffding's inequality to satisfy the assumption (see Equation (2.3)). Bernstein's inequality often provides a fast concentration rate in high-dimensional statistics. As for our problem, we notice that the assumption only holds for the optimal value function on the recovered MDP. Thus, we can utilize the variance of this optimal value function to refine the concentration bounds.

[3]Readers may skip the section at their first readings though it is important to understand the advances in the refined analysis.

Theorem 2.5 (Refined Sample Complexity of Model-based Value Iteration [Azar et al., 2013])**.** *Given a generative model $\mathcal{M}$, with the parameter*

$$n = \mathcal{O}\left(\frac{1}{\epsilon^2(1-\gamma)^3}\log\left(\frac{2\times|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right),$$

Model-based Value Iteration (see Algorithm 3) can find a sub-optimal value function $\hat{V}^*$ such that $||\hat{V}^* - V^*||_\infty \le \epsilon$ for any initial solution V_0, where $\epsilon \in (0, \frac{1}{1-\gamma}]$ is the error tolerance, with probability at least $1-\delta$. And the number of total samples used is

$$\mathcal{O}\left(\frac{|\mathcal{S}|\times|\mathcal{A}|}{\epsilon^2(1-\gamma)^3}\log\left(\frac{|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right).$$

Proof. The proof mainly refers to [Azar et al., 2013] and involves three steps. In the first step, we decompose the error bound (Lemma 2.6). Subsequently, we utilize the variance of optimal value function to apply Bernstein's inequality (Lemma 2.8). Finally but importantly, we show that the variance itself satisfies the Bellman equation, upon which we derive a tight upper for this variance (Lemma 2.10). Combining these pieces we complete the proof.

Lemma 2.6 (Elementwise bounds on $V^* - \hat{V}^*$)**.**

$$\begin{aligned} V^* - \hat{V}^* &\le \gamma\left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right)^{-1}\left[P^{\pi^*} - \hat{P}^{\pi^*}\right]V^* \\ V^* - \hat{V}^* &\ge \gamma\left(\mathbb{I} - \gamma\hat{P}^{\hat{\pi}^*}\right)^{-1}\left[P^{\pi^*} - \hat{P}^{\pi^*}\right]V^*. \end{aligned}$$

Proof. For the first inequality, using the fact that $\hat{V}^* \ge \hat{V}^{\pi^*}$ (where the inequality holds elementwise), we have

$$\begin{aligned} V^* - \hat{V}^* &\le V^* - \hat{V}^{\pi^*} \\ &= \left(\mathbb{I} - \gamma P^{\pi^*}\right)^{-1} r^{\pi^*} - \left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right)^{-1} r^{\pi^*} \quad \left(r^{\pi^*}(s) = r\left(s, \pi^*(s)\right)\right) \\ &= \left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right)^{-1}\left[\left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right) - \left(\mathbb{I} - \gamma P^{\pi^*}\right)\right]\left(\mathbb{I} - \gamma P^{\pi^*}\right)^{-1} r^{\pi^*} \\ &= \gamma\left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right)^{-1}\left[P^{\pi^*} - \hat{P}^{\pi^*}\right]\left(\mathbb{I} - \gamma P^{\pi^*}\right)^{-1} r^{\pi^*} \\ &= \gamma\left(\mathbb{I} - \gamma\hat{P}^{\pi^*}\right)^{-1}\left[P^{\pi^*} - \hat{P}^{\pi^*}\right]V^* \end{aligned}$$

Similarly, we can prove the second inequality using the fact that $V^* \ge V^{\hat{\pi}^*}$, thus the details are omitted. □

Remark. Readers may realize that this error decompose is sightly different from the Equation (2.4) that we obtained in the previous analysis. However, we note that Lemma 2.6 implies Equation (2.4) based on the facts: i) the above inequalities hold if we change the role of M and $\hat{M}$, thus RHS will be $\hat{V}^*$; ii) $||(\mathbb{I} - \gamma P^{\pi^*})^{-1}||_\infty \le 1/(1-\gamma)$. Therefore, Lemma 2.6 is more general. The benefit of this bound is that we will derive a tight bound for the estimation $[P^{\pi^*} - \hat{P}^{\pi^*}]V^*$ via considering the MDP structure hided in the term $(\mathbb{I} - \gamma P^{\pi^*})^{-1}$.

To substantiate the above idea, we introduce the variance of value function V^π as follows:

$$\sigma^2_{V^\pi}(s,a) = \gamma^2\mathbb{V}_{s'\sim p(s,a)}\left[V^\pi(s')\right], \quad \text{and} \quad \hat{\sigma}^2_{V^*}(s,a) = \gamma^2\mathbb{V}_{s'\sim\hat{p}(s,a)}\left[V^\pi(s')\right].$$

The following lemma builds the connection between $\sigma_{V^*}(s,a)$ and $\hat{\sigma}_{\hat{V}^{\pi^*}}(s,a)$. Why do we consider this connection? Recall the estimation term $[P^{\pi^*} - \hat{P}^{\pi^*}]V^*$ in Lemma 2.6. Clearly, the variance of this estimate corresponds to σ_{V^*}. However, the precondition matrix $(\mathbb{I} - \gamma P^{\pi^*})^{-1}$ depends on the structure of MDP! Thus, we need to "transfer" the variance bound.

Lemma 2.7 (Elementwise bounds on σ_{V^*})**.** *With probability at least $1-\delta$, the following relation hold separately:*

$$\sigma_{V^*} \leq \hat{\sigma}_{\hat{V}^{\pi^*}} + b_v \vec{1},$$
$$\sigma_{V^*} \leq \hat{\sigma}_{\hat{V}^{\hat{\pi}^*}} + b_v \vec{1},$$

where b_v is defined as

$$b_v = \left(\frac{18\gamma^4 \log\left(\frac{3|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^4} \right)^{1/4} + \sqrt{\frac{4\gamma^2 \log\left(\frac{6|S|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^4}}$$

See Appendix A.1 for the proof.

As mentioned before, with the result on transferred variance bound, we can bound the estimate term $[P^{\pi^*} - \hat{P}^{\pi^*}]V^*$ in Lemma 2.6 using Bernstein's inequality.

Lemma 2.8 (Elementwise bounds on $(P^{\pi^*} - \hat{P}^{\pi^*})V^*$)**.** *Define the following constants:*

$$c_{pv} = 2\log\left(\frac{2|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)$$
$$b_{pv} = \left(\frac{5\log\left(\frac{6|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n} \left(\frac{\gamma}{1-\gamma}\right)^{4/3} \right)^{3/4} + \frac{4\log\left(\frac{12|S|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^2}.$$

With probability at least $1-\delta$, we have

$$-\sqrt{\frac{c_{pv}\hat{\sigma}^2_{\hat{V}^{\pi^*}}}{n}} - b_{pv}\vec{1} \leq \gamma\left(P^{\pi^*} - \hat{P}^{\pi^*}\right)V^* \leq \sqrt{\frac{c_{pv}\hat{\sigma}^2_{\hat{V}^{\pi^*}}}{n}} + b_{pv}\vec{1}$$

Proof. For all $(s,a) \in (\mathcal{S}\times\mathcal{A})$ and $\delta \in (0,1)$, using the fact that $(P^{\pi^*} - \hat{P}^{\pi^*})V^*(s,a)$ is the sum of *i.i.d* zero-mean random variables, each of which is absolutely bounded by $1/(1-\gamma)$ and has the variance $\sigma^2_{V^*}/\gamma^2$, Bernstein's inequality [Wainwright, 2019] implies that with probability at least $1-\delta$,

$$\left(P^{\pi^*} - \hat{P}^{\pi^*}\right)V^*(s,a) \leq \sqrt{\frac{2\sigma^2_{V^*}(s,a)\log\left(\frac{2}{\delta}\right)}{n\gamma^2}} + \frac{2\log\left(\frac{2}{\delta}\right)}{3n(1-\gamma)},$$

and,

$$\left(P^{\pi^*} - \hat{P}^{\pi^*}\right)V^*(s,a) \geq -\sqrt{\frac{2\sigma^2_{V^*}(s,a)\log\left(\frac{2}{\delta}\right)}{n\gamma^2}} - \frac{2\log\left(\frac{2}{\delta}\right)}{3n(1-\gamma)}.$$

Using a union bound over all (s,a) and the estimated variance in Lemma 2.7, we have the desired result. □

According to our motivation, the tight bound is obtained by considering the combination of $(\mathbb{I}-\gamma P^{\pi^*})^{-1}$ and the variance $\hat{\sigma}_{\hat{V}^{\pi^*}}$. We will see that σ_{V^*} satisfies certain Bellman-like equation, and this property holds for any MDP. In this way, $(\mathbb{I}-\gamma P^{\pi^*})^{-1}\sigma_{V^\pi}$ looks like the concept of value function defined in the Bellman equation. Concretely, let's consider the variance of discounted rewards and define the sum of this variance as

$$\Sigma^\pi(s,a) = \mathbb{E}\left[\left\{\sum_{t\geq 0}\gamma^t r\left(s_t,a_t\right) - Q^\pi(s,a)\right\}^2 \mid s_0 = s, a_0 = a\right].$$

Lemma 2.9 (Bellman-like variance)**.** Σ^π *satisfies the following Bellman equation:*

$$\Sigma^\pi = \sigma^2_{V^\pi} + \gamma^2 P^\pi \Sigma^\pi.$$

Proof. For all (s,a), we have (for simplicity, we will omit the condition and the subscript of the expectation when there is no ambiguity):

$$
\begin{aligned}
\Sigma^{\pi}(s,a) &= \mathbb{E}\left[\left\{\sum_{t\geq 0}\gamma^t r\left(s_t,a_t\right)-Q^{\pi}(s,a)\right\}^2 \Big| s_0=s,a_0=a\right] \\
&= \mathbb{E}_{(s_1,a_1)}\mathbb{E}\left[\left\{\sum_{t\geq 1}^{\pi}\gamma^t r\left(s_t,a_t\right)-\gamma Q^{\pi}\left(s_1,a_1\right)-\left(Q^{\pi}(s,a)-r(s,a)-\gamma Q^{\pi}\left(s_1,a_1\right)\right)\right\}^2 \Big|\right] \\
&= \gamma^2\mathbb{E}_{(s_1,a_1)}\mathbb{E}\left[\left|\sum_{t\geq 1}^{t}\gamma^{t-1} r\left(s_t,a_t\right)-Q^{\pi}\left(s_1,a_1\right)\right|^2\right] \\
&\quad -2\mathbb{E}_{(s_1,a_1)}\left[\left(Q^{\pi}(s,a)-r(s,a)-\gamma Q^{\pi}\left(s_1,a_1\right)\right)\mathbb{E}\left(\sum_{t\geq 1}\gamma^t r\left(s_t,a_t\right)-\gamma Q^{\pi}\left(s_1,a_1\right)\mid\left(s_1,a_1\right)\right)\right] \\
&\quad +\mathbb{E}_{(s_1,a_1)}\left[\left|Q^{\pi}(s,a)-r(s,a)-\gamma Q^{\pi}\left(s_1,a_1\right)\right|^2\right] \\
&\overset{(i)}{=} \gamma^2\mathbb{E}_{(s_1,a_1)}\mathbb{E}\left[\left|\sum_{t\geq 1}\gamma^{t-1} r\left(s_t,a_t\right)-Q^{\pi}\left(s_1,a_1\right)\right|^2\right]+\gamma^2\mathbb{V}_{s_1\sim p(\cdot|s,a)}\left[Q^{\pi}\left(s_1,\pi\left(s_1\right)\right)\right] \\
&= \gamma^2\left[P^{\pi}\Sigma^{\pi}\right](s,a)+\sigma^2_{V^{\pi}}(s,a).
\end{aligned}
$$

where step (i) relies on $\mathbb{E}\left[\sum_{t\geq 1}\gamma^t r\left(s_t,a_t\right)-\gamma Q^{\pi}\left(s_1,a_1\right)\mid\left(s_1,a_1\right)\right]=0$. □

Lemma 2.10 (Tight bounds on the variance of value function)**.** *Based on Lemma 2.9, we can show that*

$$
\begin{aligned}
\left\|\left(\mathbb{I}-\gamma^2 P^{\pi}\right)^{-1}\sigma^2_{V^{\pi}}\right\|_{\infty} &= \|\Sigma^{\pi}\|_{\infty}\leq\frac{1}{(1-\gamma)^2} \\
\left\|\left(\mathbb{I}-\gamma P^{\pi}\right)^{-1}\sigma_{V^{\pi}}\right\|_{\infty} &\leq 2\|\sqrt{\frac{1}{1-\gamma}\Sigma^{\pi}}\|_{\infty}\leq\frac{2}{(1-\gamma)^{1.5}}
\end{aligned}
$$

See Appendix A.2 for the proof.

Plugging the variance estimate in Lemma 2.10 to Lemma 2.8, based on the error decomposition in Lemma 2.6, we can conclude that with probability at least $1-\delta$,

$$||V^*-\hat{V}^*||_{\infty}\leq b,$$

where b is defined as

$$
b=\sqrt{\frac{4\log\left(\frac{2|S|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^3}}+\left(\frac{5\log\left(\frac{6|S|\times|\mathcal{A}|}{\delta}\right)}{n}\left(\frac{\gamma}{(1-\gamma)^2}\right)^{4/3}\right)^{3/4}+\frac{4\log\left(\frac{12|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^3}.
$$

We can see that the first term dominates, and if we are provided

$$
T=\mathcal{O}\left(\frac{|\mathcal{S}|\times|\mathcal{A}|}{\epsilon^2(1-\gamma)^3}\log\left(\frac{2|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)\right)
$$

samples to construct the model, the learned optimal value function on the recovered MDP is ϵ-optimal on the original MDP. □

2.3 Lower Bound on Generative Model

We may ask the question: can we further design better algorithms for the setting of the generative model? To answer this question, we take an information-theoretically perspective, that is, how many the minimal samples/information do an intelligent algorithm need to solve for all MDPs? Thus, we need a lower bound, which works for all algorithms. If the upper bound by some smart algorithm matches the lower bound (the difference of logarithmic or constant terms can be ignored), we conclude that this algorithm is minimax optimal and cannot be improved.

Theorem 2.11 (Lower bound on generative model)**.** *There exist some constants $\epsilon_0, \delta_0, c_1, c_2$ and a class MDPs $\mathbb{M}$, such that for all $\epsilon \in (0, \epsilon_0), \delta \in (0, \delta_0)$, and every (ϵ, δ)-correct RL algorithm on the class of MDPs $\mathbb{M}$, the total number of state-transition samples needs to be at least*

$$T = \left\lceil \frac{|\mathcal{S}| \times |\mathcal{A}|}{c_1 \epsilon^2 (1-\gamma)^3} \log\left(\frac{|\mathcal{S}| \times |\mathcal{A}|}{c_2 \delta}\right) \right\rceil.$$

See Appendix A.3 for the proof. Firstly, we clearly see that the upper bound of *Model-based Value Iteration* (see Theorem 2.5) matches this lower bound. Hence, *Model-based Value Iteration* is minimax optimal.

3 Variance-reduced Value Iteration

In Section 2.1, we have analyzed the simple model-free algorithm *Phased Value Iteration* and we see that the sample complexity of this algorithm is not minimax optimal. In this section, we attempt to improve this algorithm with a technique called **variance-reduction**. Our key insights come from two aspects: 1) the convergence rate of model-free algorithm is usually linear, however, we pose the uniform accuracy on the sampling-based Bellman update in the previous analysis, which clearly does not harness the geometric structure of this problem and correspondingly wastes lots of samples when $\hat{V}_t$ is far from V^*; 2) we tackle the estimation problem with the range of $[0, 1/(1-\gamma)]$ in the prior proof, however, we can introduce some reference variable to shrink the estimation range; that is, we only need to estimate the residual part $V^* - \hat{V}_t$, which decays over iterations and in this way we may not require the original $\mathcal{O}(1/(1-\gamma)^4)$ samples.

Algorithm 4 Approximate Value Operator: $ApxVal(V, \tilde{V}, \mathcal{X}, \epsilon, \delta)$

Input: current value $V \in \mathbb{R}^{|\mathcal{S}|}$, and reference-point $\tilde{V} \in \mathbb{R}^{|\mathcal{S}|}$.
Input: precomputed offset $\mathcal{X} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}$ such that $|\mathcal{X}(s,a) - \mathbb{E}_{s'\sim p(\cdot|s,a)}[\tilde{V}(s')]| \leq \epsilon$ for all (s,a).
Input: desired accuracy $\epsilon \in (0,1)$ and failure probability $\delta \in (0,1)$.

Set
$$n = \left\lceil \frac{2||V - \tilde{V}||_\infty^2}{\epsilon^2} \log\left(\frac{2}{\delta}\right) \right\rceil.$$
Initialize variables $Z_t \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}$, $\bar{V} \in \mathbb{R}^{|\mathcal{S}|}$ and $\pi \in \mathbb{R}^{|\mathcal{S}|}$.
for each state $s \in \mathcal{S}$ **do**
  for each action $a \in \mathcal{A}$ **do**
    Sample n next states $\{s_i'\}_{i=1}^n$ by calling $\mathcal{M}$.
    Set $Z(s,a) = r(s,a) + \gamma\left(\mathcal{X}(s,a) + \sum_{i=1}^n \frac{1}{n}[V(s_i') - \tilde{V}(s_i')]\right)$.
  end for
  Set $\bar{V}(s) = \max_{a\in\mathcal{A}} Z(s,a)$, and $\pi(s) = \arg\max_{a\in\mathcal{A}} Z(s,a)$.
end for

Output: $(\bar{V}, \pi)$.

To understand the technique of variance-reduction, we first introduce a reference-point based value estimation algorithm. The ingredient is that we can introduce a new estimator

$$V^\dagger = \mathbb{E}[\tilde{V}] + V - \tilde{V},$$

where $\tilde{V}$ is some reference-point and the expectation is taken over the transition probability $p(\cdot|s,a)$ that is conditioned on some (s,a). It is easy to verify that this estimator is unbiased, that is, $\mathbb{E}[V^\dagger] = \mathbb{E}[V]$. The benefit of this estimator is that we only need to estimate the term $V - \tilde{V}$, whose range is small than the original one under some condition. Unfortunately, we may not easily get $\mathbb{E}[\tilde{V}]$, which also requires the knowledge of transition probability. To tackle this issue, we again use the sampling-based estimator and introduce $\hat{V} = \sum_{i=1}^n \frac{1}{n}\tilde{V}(s_i') \approx \mathbb{E}[\tilde{V}]$. The estimation algorithm based on the above idea is presented in the following algorithms.

3.1 Sublinear Randomized Value Iteration

In this part, we present the first algorithm that uses the variance-reduction technique to improve the sample complexity of model-free methods. Though this algorithm called *Sublinear Randomized Value Iteration* does not match the lower bound, it's a good example to understand the intuition regarding the improvement. Unless stated, the following algorithm design and corresponding analysis are from [Sidford et al., 2018].

Lemma 3.1 (Quality of Approximate Value Operator)**.** *With probability at least $1-\delta$, the output of Approximate Value Operator (see Algorithm 4) satisfies that*

$$||\bar{V} - \mathcal{T}(V)||_\infty \leq 2\gamma\epsilon.$$

Proof. Given the condition, we have that for all (s, a),

$$\left| \mathbb{E}_{s'\sim p(\cdot|s,a)} \left[V(s') - \tilde{V}(s') \right] - \sum_{i=1}^{n} \frac{1}{n} \left[V(s'_i) - \tilde{V}(s'_i) \right] \right| \leq \epsilon,$$

with probability at least $1-\delta$. Then, like what we have done in the proof of *Phased Value Iteration*, we have for each $s \in \mathcal{S}$,

$$\begin{aligned}
&\left|\bar{V}(s) - \mathcal{T}(V)(s)\right| \\
&= \left| \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')] \right\} - \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \left[\mathcal{X}(s,a) + \sum_{i=1}^{n} \frac{1}{n} \left[V(s'_i) - \tilde{V}(s'_i) \right] \right] \right\} \right| \\
&\leq \max_{a\in\mathcal{A}} \left| \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')] \right\} - \left\{ r(s,a) + \gamma \left[\mathcal{X}(s,a) + \sum_{i=1}^{n} \frac{1}{n} \left[V(s'_i) - \tilde{V}(s'_i) \right] \right] \right\} \right| \\
&= \gamma \max_{a\in\mathcal{A}} \left| \left(\mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')] - \sum_{i=1}^{n} \frac{1}{n} V(s'_i) \right) + \left(\sum_{i=1}^{n} \frac{1}{n} \tilde{V}(s'_i) - \mathcal{X}(s,a) \right) \right| \\
&\leq 2\gamma\epsilon.
\end{aligned}$$

□

Remark. *Approximate Value Operator* is very similar to the naive sampling-based estimation operator. However, the former is sightly worse than the latter because the additional error of reference-point. One should not worry because this adverse effect is only constant-level. Based on *Approximate Value Operator*, we introduce a new value iteration method, which is outlined in Algorithm 5.

Algorithm 5 Sampled Randomized Value Iteration: *SampledRandomizedVI*$(V_0, L, \epsilon, \delta)$

Input: initial value V_0 and number of iterations $L > 0$
Input: target accuracy $\epsilon > 0$ and failure probability $\delta \in (0, 1)$
1: Sample n samples to obtain approximate offsets: $\mathcal{X} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}$ with $|\mathcal{X}(s,a) - \mathbb{E}_{s'\sim p(\cdot|s,a)}[V_0(s')]| \leq \epsilon$ for all (s, a).

$$n = \left\lceil \frac{2||V_0||_\infty^2}{\epsilon^2} \log\left(\frac{2L}{\delta}\right) \right\rceil.$$

2: **for** each round $\ell = 1, 2, \cdots L$ **do**
3: $\quad (V_\ell, \pi_\ell) = ApxVal(V_{\ell-1}, V_0, \mathcal{X}, \epsilon, \delta/(2L))$.
4: **end for**
Output: (V_L, π_L).

Lemma 3.2 (Sample Complexity of Sampled Randomized Value Iteration)**.** *An invocation of Sample Complexity of Sampled Randomized Value Iteration (see Algorithm 5) requires*

$$\mathcal{O}\left(|\mathcal{S}| \times |\mathcal{A}| \left[\frac{1}{(1-\gamma)^2\epsilon^2} + L \left(\frac{||V_0 - V^*||_\infty^2}{\epsilon^2} + \frac{1}{(1-\gamma)^2} \right) \right] \log\left(\frac{|\mathcal{S}| \times |\mathcal{A}| \times L}{\delta} \right) \right)$$

samples. Furthermore, if we set

$$L = \left\lceil \frac{1}{1-\gamma} \log\left(\frac{2||V_0 - V^*||_\infty}{(1-\gamma)\epsilon}\right)\right\rceil,$$

the output of Sample Complexity of Sampled Randomized Value Iteration satisfies that $||V_L - V^*||_\infty \leq \epsilon$*, and correspondingly the number of required samples becomes*

$$\mathcal{O}\left(\frac{|\mathcal{S}| \times |\mathcal{A}|}{\epsilon^2(1-\gamma)^5} \log\left(\frac{|\mathcal{S}| \times |\mathcal{A}| \times L}{\delta}\right)\right)$$

Proof. The sample complexity is computed as follows. To compute the offset, according to Hoeffding's inequality, we require $\mathcal{O}(\frac{1}{(1-\gamma)^2\epsilon^2})$ samples because $||V_0||_\infty \leq 1/(1-\gamma)$. For the loop, the estimation range is bounded as

$$\begin{aligned}
\|V_\ell - V_0\|_\infty^2 &\leq \|V_\ell - V^* + V^* - V_0\|_\infty^2 \\
&\leq (||V_\ell - V^*||_\infty + ||V^* - V_0||_\infty)^2 \\
&\overset{(i)}{\leq} 2||V_\ell - V^*||_\infty^2 + 2||V^* - V_0||_\infty^2 \\
&\overset{(ii)}{\leq} 2\left(\frac{2\gamma\epsilon}{1-\gamma} + ||V_0 - V^*||_\infty\right)^2 + 2||V^* - V_0||_\infty^2 \\
&\leq \frac{8\epsilon^2}{(1-\gamma)^2} + 8||V_0 - V^*||_\infty^2,
\end{aligned} \tag{3.1}$$

where step (i) uses the inequality that $(a+b)^2 \leq 2a^2 + 2b^2$, and step (ii) is based on the error bound shown in the following Equation (3.2). Therefore, the loop takes $\mathcal{O}\left(|\mathcal{S}| \times |\mathcal{A}| \times L\left(\frac{1}{(1-\gamma)^2} + \frac{||V_0 - V^*||_\infty^2}{\epsilon^2}\right)\right)$ samples. Adding the number of offset and the loop yields the result.

From Lemma 3.1, we see that the approximate value operator only incurs $2\gamma\epsilon$ error compared with the population-based Bellman operator. Thus, define $V_\ell^\sharp$ as the ℓ-th iteration result of *Value Iteration*, we can derive the error bound of *Sampled Randomized Value Iteration* as follows:

$$\begin{aligned}
\|V_\ell - V^*\|_\infty &\leq \left\|V_\ell - V_\ell^\sharp\right\|_\infty + \left\|V_\ell^\sharp - V^*\right\|_\infty \\
&= ||V_\ell - \mathcal{T}(V_{\ell-1}) + \mathcal{T}(V_{\ell-1}) - V_\ell^\sharp|| + \left\|V_\ell^\sharp - V^*\right\|_\infty \\
&\leq 2\gamma\epsilon + \gamma\left\|V_{\ell-1} - V_{\ell-1}^\sharp\right\|_\infty + \gamma^\ell \|V_0 - V^*\|_\infty \\
&\leq \frac{2\gamma\epsilon}{1-\gamma} + \gamma^\ell \|V_0 - V^*\|_\infty .
\end{aligned} \tag{3.2}$$

Correspondingly, if

$$L \geq \left\lceil \frac{1}{1-\gamma} \log\left(\frac{||V_0 - V^*||_\infty}{2\gamma\epsilon}\right)\right\rceil,$$

then

$$||V_L - V^*||_\infty \leq (4\gamma\epsilon)/(1-\gamma). \tag{3.3}$$

By simple algebraic transformation, we get the desired result. □

Remark. Unfortunately, *Sampled Randomized Value Iteration* itself does not improve the sample-efficiency. In the following, we present a carefully designed algorithm based on *Sampled Randomized Value Iteration* to refined the sample complexity. The key insight is that we adaptively fit the estimation range and the estimation error (the blue part in Lemma 3.2).

Algorithm 6 Sublinear Randomized Value Iteration: *SublinearRandomizedVI*(ϵ, δ)

Input: desired precision ϵ and failure probability $\delta \in (0, 1)$.

Set $K = \left\lceil \log_2 \left(\frac{1}{\epsilon(1-\gamma)} \right) \right\rceil$, and $L = \left\lceil \frac{1}{1-\gamma} \log \left(\frac{4}{1-\gamma} \right) \right\rceil$

Set $V_0 = \vec{0}$ and $\epsilon_0 = \frac{1}{1-\gamma}$.

for each iteration $k = 1, 2, \cdots, K$ **do**

  Set $\epsilon_k = \frac{1}{2}\epsilon_{k-1} = \frac{1}{2^k(1-\gamma)}$.

  $(V_k, \pi_k) =$ *SampledRandomizedVI*$(V_{k-1}, L, (1-\gamma)\epsilon_k/(4\gamma), \delta/K)$.

end for

Output: (V_K, π_K).

Lemma 3.3 (Quality of Sublinear Randomized Value Iteration)**.** *With probability at least $1-\delta$, the iterator V_k of Sublinear Randomized Value Iteration (see Algorithm 6) satisfies that $||V_k - V^*||_\infty \leq \epsilon_k$ for all $k \in [0, K]$. And considering the given outer loop iteration number K, V_k is ϵ-optimal.*

Proof. We use the inductive reasoning. For the base case, $||V_0 - V^*|| \leq 1/(1-\gamma) = \epsilon_0$. For the inductive case, we assume that $||V_k - V^*||_\infty \leq \epsilon_k$ for some $k \geq 0$, and we want to prove that $||V_{k+1} - V^*||_\infty \leq \epsilon_{k+1} = 1/(2\epsilon_k)$. According to Lemma 3.2, we only need to ensure that the inner iteration number is large enough. Actually, we need to guarantee that

$$\begin{aligned} L &\geq \left\lceil \frac{1}{1-\gamma} \log \left(\frac{||V_k - V^*||_\infty}{2\gamma} \times \frac{4\gamma}{(1-\gamma)\epsilon_k} \right) \right\rceil \\ &= \left\lceil \frac{1}{1-\gamma} \log \left(\frac{2||V_{k-1} - V^*||_\infty}{(1-\gamma)\epsilon_k} \right) \right\rceil. \end{aligned}$$

Based on our assumption that $||V_{k-1} - V^*||_\infty \leq \epsilon_{k-1} = 2\epsilon_k$, we recover the value chosen in the algorithm design. Finally, we can check that given the inner iteration number, the output satisfies that

$$||V_k - V^*||_\infty \leq \frac{4\gamma}{1-\gamma} \times \frac{(1-\gamma)\epsilon_k}{4\gamma} = \epsilon_k.$$

Finally, notice that the given outer loop iteration number guarantees that V_K is ϵ-optimal. □

Remark. We can see that the desired accuracy of inner loop in Algorithm 6 adaptively fits the estimation range $||V_k - V^*||$. In this way, the sample complexity of model-free algorithm can be improved.

Theorem 3.4 (Sample Complexity of Sublinear Randomized Value Iteration [Sidford et al., 2018])**.** *In invocation of Sublinear Randomized Value Iteration (see Algorithm 6) requires*

$$\tilde{\mathcal{O}} \left(|\mathcal{S}| \times |\mathcal{A}| \left(\frac{1}{\epsilon^2(1-\gamma)^4} + \frac{1}{(1-\gamma)^3} \right) \log \left(\frac{1}{\delta} \right) \right)$$

samples to obtain an ϵ-optimal value function with probability at least $1-\delta$, where $\epsilon \in (0, \frac{1}{1-\gamma}]$.

Proof. The desired result is mainly based on Equation (3.1) by replacing ϵ with $(1-\gamma)\epsilon_k/(4\gamma)$ and L with $\left\lceil \frac{1}{1-\gamma} \log \left(\frac{4}{1-\gamma} \right) \right\rceil$. And note that iteration number K is logarithmic term, which is ignored in the notation of $\tilde{\mathcal{O}}(\cdot)$. □

Remark. The sample complexity of *Sublinear Randomized Value Iteration* is clearly better than the one of *Phased Value Iteration* (see Theorem 2.3). However, it is not clear that which term ($\tilde{\mathcal{O}}(1/(\epsilon^2(1-\gamma)^4))$ or $\tilde{\mathcal{O}}(1/(1-\gamma)^3)$) will dominate in practice.

3.2 Monotonic Sublinear Randomized Value Iteration

In all previous sections, we always consider algorithms that output an ϵ-optimal value function. However, we do not deep think about how to obtain an ϵ-optimal policy. At first glance, it may seem that an ϵ-optimal value function implies an ϵ-optimal policy. However, this conjecture is not correct. The reason is that the induced greedy policy may make a mistake from some timestep (because the value function is not perfect) and subsequently make another mistake at a new state; in this way, the decision error compounds. This phenomenon is summarized in the following lemma.

Lemma 3.5 (Sub-optimality of Induced Greedy Policy)**.** *Suppose the value function V is ϵ-optimal, that is, $\|V - V^*\|_\infty \le \epsilon$, and π is the induced policy, that is, $\pi(s) = \arg\max_{a\in\mathcal{A}} r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')]$, then we have*

$$\|V_\pi - V^*\|_\infty \le \frac{2\gamma}{1-\gamma}\epsilon.$$

Proof. For any policy deterministic $\pi \in \mathcal{A}^{|\mathcal{S}|}$ and value function $V \in \mathbb{R}^{|\mathcal{S}|}$, let $P_\pi V(s) = \mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')]$, and $r_\pi(s) = r(s,\pi(s))$.

$$\begin{aligned}
&\|V_\pi - V^*\|_\infty \\
&\le \|(r_\pi + \gamma P_\pi V_\pi) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty \\
&\le \|(r_\pi + \gamma P_\pi V_\pi) - (r_\pi + \gamma P_\pi V^*)\|_\infty + \|(r_\pi + \gamma P_\pi V^*) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty \\
&\le \gamma \|V_\pi - V^*\|_\infty + \|(r_\pi + \gamma P_\pi V^*) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty ,
\end{aligned}$$

then, we have

$$\|V_\pi - V^*\|_\infty \le \frac{1}{1-\gamma} \|(r_\pi + \gamma P_\pi V^*) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty .$$

For the RHS, we future have

$$\begin{aligned}
&\|(r_\pi + \gamma P_\pi V^*) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty \\
&\le \|(r_\pi + \gamma P_\pi V^*) - (r_\pi + \gamma P_\pi V) + (r_\pi + \gamma P_\pi V) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty \\
&\le \gamma \|V - V^*\|_\infty + \|(r_\pi + \gamma P_\pi V) - (r_{\pi^*} + \gamma P_{\pi^*} V^*)\|_\infty \\
&= \gamma \|V - V^*\|_\infty + \max_{s\in\mathcal{S}} \left| \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)}[V(s')] \right\} - \max_{a\in\mathcal{A}} \left\{ r(s,a) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,a)}[V^*(s')] \right\} \right| \qquad (3.4) \\
&\overset{(i)}{\le} \gamma \|V - V^*\|_\infty + \gamma \max_{s\in\mathcal{S}} |V(s) - V^*(s)| \\
&\le 2\gamma \|V - V^*\|_\infty ,
\end{aligned}$$

where step (i) is based on the maximum inequality, which we have used many times (e.g., ref to Equation (2.2)). Finally, we conclude that

$$\|V_\pi - V^*\|_\infty \le \frac{2\gamma}{1-\gamma}\epsilon.$$

□

Remark. Lemma 3.5 implies that the quality of induced greedy policy by the imperfect value function is worse. By reviewing the previous bounds (ref to Theorem 2.3, 2.4, 2.5 and 3.4), we could find that these naive methods lead to at least more $1/(1-\gamma)^2$ complexity for induced policies. And we may want to know that can we escape the curse of the horizon? If so, how can we modify our algorithms to get a better policy?

These questions are first answered in [Sidford et al., 2018]. In the next, we present how to improve the design of Algorithm 6 to obtain a better policy than the naive versions. First of all, we note that Lemma 3.2

is built on the assumption that transition P is know so that we can accurately compute the greedy policy. However, we can only obtain a greedy policy by sampling. Thus, we need to build new bounds for the latter. To start with, we note that for any iteration $\ell = 1, 2, \cdots, L$ in Algorithm 5, we have

$$\|V_{\pi_\ell} - V^*\|_\infty \leq \|V_{\pi_\ell} - V_\ell\|_\infty + \|V_\ell - V^*\|_\infty . \tag{3.5}$$

Actually, we have built the upper bound for the second term of RHS (ref to Equation (3.2)). Thus, we only need to focus on the first term. Before proceeding, let's guess which term dominates? Somewhat surprisingly, the first term incurs $\mathcal{O}((1-\gamma)^{-2})$ while the last term only incurs $\mathcal{O}((1-\gamma)^{-1})$. The reason is similar to what we have explained for Lemma 3.5. Over iterations, the value function only accumulates linear errors while the induced policy additionally suffers more errors. In the following, we rigorously state this result. However, we cannot directly apply previous threads because π_ℓ is no longer to be the greedy policy under the true environments (ref to the 4-th line of Equation (3.4)). To begin with, for any vector $V \in \mathbb{R}^{|\mathcal{S}|}$ and any deterministic policy $\pi \in \mathcal{A}^{|\mathcal{S}|}$, we define

$$\mathcal{T}_\pi(V)(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,\pi(s))}[V(s')].$$

By triangle inequality, we have

$$\|V_{\pi_\ell} - V_\ell\|_\infty \leq \|V_{\pi_\ell} - \mathcal{T}_{\pi_\ell} V_{\ell-1}\|_\infty + \|\mathcal{T}_{\pi_\ell} V_{\ell-1} - V_\ell\|_\infty.$$

We can further separately upper bound the above two terms using the following relations. We first upper bound first term with Lemma 3.6.

Lemma 3.6. *For any vector $V \in \mathbb{R}^{|\mathcal{S}|}$ and any deterministic policy $\pi \in \mathcal{A}^{|\mathcal{S}|}$, we have*

$$\|V_\pi - \mathcal{T}_\pi V\|_\infty \leq \frac{\gamma}{1-\gamma} \|V - \mathcal{T}_\pi V\|_\infty .$$

See Appendix B.1 for the proof. With Lemma 3.6, we have

$$\begin{aligned}
\|V_{\pi_\ell} - V_\ell\|_\infty &\leq \frac{\gamma}{1-\gamma} \|V_{\ell-1} - \mathcal{T}_\pi V_{\ell-1}\|_\infty + \|\mathcal{T}_{\pi_\ell} V_{\ell-1} - V_\ell\|_\infty \\
&\leq \frac{\gamma}{1-\gamma} \|V_{\ell-1} - V_\ell\|_\infty + \frac{1}{1-\gamma} \|\mathcal{T}_{\pi_\ell} V_{\ell-1} - V_\ell\|_\infty \\
&\leq \frac{\gamma}{1-\gamma} (\|V_\ell - V^*\|_\infty + \|V_{\ell-1} - V^*\|_\infty) + \frac{1}{1-\gamma} \|\mathcal{T}_{\pi_\ell} V_{\ell-1} - V_\ell\|_\infty.
\end{aligned}$$

Back to Equation (3.5), when the iteration number is large enough, e.g., if

$$\ell = L = \left\lceil \frac{1}{1-\gamma} \log \left(\frac{2||V_0 - V^*||_\infty}{(1-\gamma)\epsilon} \right) \right\rceil ,$$

we have

$$\begin{aligned}
\|V_{\pi_L} - V^*\|_\infty &\leq \frac{1}{1-\gamma} \|V_L - V^*\|_\infty + \frac{\gamma}{1-\gamma} \|V_{L-1} - V^*\|_\infty + \frac{1}{1-\gamma} \|\mathcal{T}_{\pi_L} V_{L-1} - V_\ell\|_\infty \\
&\leq \frac{4\epsilon\gamma}{(1-\gamma)^2} + \frac{4\epsilon\gamma}{(1-\gamma)^2} + \frac{1}{1-\gamma} \|\mathcal{T}_{\pi_L} V_{L-1} - V_L\|_\infty,
\end{aligned} \tag{3.6}$$

where the last inequality follows the previous results (ref to Equation (3.3)). Now, we deal with the last term.

Lemma 3.7. *With probability at least $1-\delta$, the output of Approximate Value Operator (see Algorithm 4) satisfies that*

$$\left\|\mathcal{T}_\pi V - \bar{V}\right\|_\infty \leq 2\gamma\epsilon.$$

Proof. At first glance, the meaning of this error term is obscure. Actually, according to the definition, we have

$$\mathcal{T}_\pi(V)(s) = r(s,\pi(s)) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V(s')],$$
$$\bar{V} = r(s,\pi(s)) + \gamma\left(\mathcal{X}(s,\pi(s)) + \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[V(s') - \tilde{V}(s')]\right).$$

Therefore, they are mainly different in the transition probability.

$$\begin{aligned}
&\left\|\mathcal{T}_\pi V - \bar{V}\right\|_\infty \\
&= \gamma \max_{s\in\mathcal{S}} \left|\mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V(s')] - \left(\mathcal{X}(s,\pi(s)) + \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[V(s') - \tilde{V}(s')]\right)\right| \\
&\le \gamma \max_{s\in\mathcal{S}} \left\{\left|\mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V(s')] - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[V(s')]\right| + \left|\mathcal{X}(s,\pi(s)) - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[\tilde{V}(s')]\right|\right\} \\
&\le \gamma \max_{s\in\mathcal{S}} \left\{\left|\mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V(s')] - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[V(s')]\right|\right\} + \gamma \max_{s\in\mathcal{S}} \left\{\left|\mathcal{X}(s,\pi(s)) - \mathbb{E}_{s'\sim \hat{p}(\cdot|s,\pi(s))}[\tilde{V}(s')]\right|\right\} \\
&\le 2\gamma\epsilon.
\end{aligned}$$

□

Now we have solved the blue part and we conclude our result with the following lemma.

Lemma 3.8 (Quality of Policy of Sampled Randomized Value Iteration)**.** *With probability at least* $1-\delta$*, the output policy of Sampled Randomized Value Iteration (see Algorithm 5) satisfies that*

$$\|V_{\pi_L} - V^*\|_\infty \le \frac{16\epsilon\gamma}{(1-\gamma)^2}.$$

Remark. The proof is completed by bounding the error term in Equation (3.6) with Lemma 3.7. We notice that Lemma 3.8 substantially reveals the sub-optimality of the induced policy for a randomized algorithm. Different from Lemma 3.5, the greedy policy is computed by sampling.

In the next, we present the monotonic algorithm that guarantees $V \le V_\pi \le V^*$ element-wise. In this way, V_π is restricted in the range of $[V, V^\pi]$, thus its sub-optimality is the same order with the one of V. Therefore, we avoid the additional $(1-\gamma)$ error for the policy value.

Algorithm 7 Monotonic Random Value Operator: *ApxMonVal*$(V,\pi,\tilde{V},\mathcal{X},\epsilon,\delta)$

Input: current value $V \in \mathbb{R}^{|\mathcal{S}|}$, and reference-point $\tilde{V} \in \mathbb{R}^{|\mathcal{S}|}$, policy $\pi \in \mathbb{R}^{|\mathcal{S}|}$ with $\mathcal{T}_\pi(V) \ge V$.
Input: precomputed offset $\mathcal{X} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}$ such that $|\mathcal{X}(s,a) - \mathbb{E}_{s'\sim p(\cdot|s,a)}[\tilde{V}(s')]| \le \epsilon$ for all (s,a).
Input: desired accuracy $\epsilon \in (0,1)$ and failure probability $\delta \in (0,1)$.

```
(q, w) = ApxVal(V, Ṽ, X, ε, δ).
for all each s ∈ S do
    if q(s) − 2γε > V(s) then
        V†(s) = q(s) − 2γε.
        π†(s) = w(s).
    else
        V†(s) = V(s).
        π†(s) = π(s).
    end if
end for
```

Output: $(V^\dagger, \pi^\dagger)$.

Lemma 3.9 (Monotonicity Properties)**.** *With probability at least $1-\delta$, the output of Monotonic Random Value Operator (see Algorithm 7) satisfies*

$$V \le V^\dagger \le \mathcal{T}_{\pi^\dagger}(V) \le \mathcal{T}_{\pi^\dagger}(V^\dagger),$$

where $\le$ holds elementwise. Importantly, the value function of $\pi^\dagger$ satisfies that

$$V \le V_{\pi^\dagger} \le V^*.$$

Proof. By design, $V \le V^\dagger$ holds trivially. By the contractility of $\mathcal{T}_{\pi^\dagger}$ (ref to Lemma B.1), $\mathcal{T}_{\pi^\dagger}(V) \le \mathcal{T}_{\pi^\dagger}(V^\dagger)$ also holds. In the following, we only need to show $V^\dagger \le \mathcal{T}_{\pi^\dagger}(V)$.

First of all, note that by Lemma 3.7, with probability at least $1-\delta$, we have $||q - \mathcal{T}_w(V)||_\infty \le 2\gamma\epsilon$. Therefore

$$-2\gamma\epsilon \le \mathcal{T}_w(V)(s) - q(s) \le 2\gamma\epsilon, \text{ for all } s \in \mathcal{S}.$$

In the case where $q(s) - 2\gamma\epsilon \ge V(s)$, we have

$$V^\dagger(s) = q(s) - 2\gamma\epsilon \le \mathcal{T}_w(V)(s) = \mathcal{T}_{\pi^\dagger}(V)(s),$$

and in the case where $q(s) - 2\gamma\epsilon \le V(s)$, we have

$$V^\dagger(s) = V(s) \le \mathcal{T}_\pi(V)(s) = \mathcal{T}_{\pi^\dagger}(V)(s),$$

by assumptions. Therefore, in either case we have that $V^\dagger \le \mathcal{T}_{\pi^\dagger}(V)$.

Finally, we demonstrate the monotonicity of value functions. Again, by the contractility of $\mathcal{T}_{\pi^\dagger}$, we know that $\mathcal{T}_{\pi^\dagger}(V^\dagger) \le \mathcal{T}_{\pi^\dagger}(\mathcal{T}_{\pi^\dagger}V^\dagger)$. And repeatedly applying this inequality, we have

$$V^\dagger \le \mathcal{T}_{\pi^\dagger}(V) \le \mathcal{T}^2_{\pi^\dagger}(V^\dagger) \le \cdots \le \mathcal{T}^\infty_{\pi^\dagger}(V^\dagger) = V_{\pi^\dagger}.$$

□

Remark. By monotonicity of *ApxMonVal*, the value function of the induced policy is squeezed, therefore its sub-optimality is almost the same with the one of $V^\dagger$. However, by the "decorator", we need to ensure the basic property of *ApxVal* does not change (too much).

Lemma 3.10 (Quality of Monotonic Random Value Operator)**.** *With probability at least $1-\delta$, the output of Monotonic Random Value Operator satisfies that*

$$\left\|V^\dagger - \mathcal{T}V\right\|_\infty \le 4\gamma\epsilon, \text{ and } \left\|V^\dagger - \mathcal{T}_{\pi^\dagger}V\right\|_\infty \le 4\gamma\epsilon.$$

Proof. By Lemma 3.1 and Lemma 3.7, we have that with probability at least $1-\delta$, $||q - \mathcal{T}V||_\infty \le 2\gamma\epsilon$ and $||q - \mathcal{T}_w V||_\infty \le 2\gamma\epsilon$.

Now in the case where $q(s) - 2\gamma\epsilon > V(s)$, then

$$\begin{aligned}\left|V^\dagger(s) - \mathcal{T}V(s)\right| &= |q(s) - 2\gamma\epsilon - \mathcal{T}V(s)| \\ &\le \|q - \mathcal{T}V\|_\infty + 2\gamma\epsilon \\ &= 4\gamma\epsilon.\end{aligned}$$

Similarly,

$$\begin{aligned}\left|V^\dagger(s) - \mathcal{T}_{\pi^\dagger}V(s)\right| &= |q(s) - 2\gamma\epsilon - \mathcal{T}_{\pi^\dagger}V(s)| \\ &\le \|q - \mathcal{T}_{\pi^\dagger}V\|_\infty + 2\gamma\epsilon \\ &= 4\gamma\epsilon.\end{aligned}$$

On the other hand, in the case where $q(s) - 2\gamma\epsilon \leq V(s)$, we have

$$\begin{aligned}\left|V^\dagger(s) - \mathcal{T}V(s)\right| &= |V(s) - \mathcal{T}V(s)| \\ &\overset{(i)}{=} \mathcal{T}V(s) - q(s) + q(s) - V(s) \\ &\leq 2\gamma\epsilon + 2\gamma\epsilon \\ &= 4\gamma\epsilon,\end{aligned}$$

where step (i) follows our assumption that $V \leq \mathcal{T}_\pi V \leq \mathcal{T}V$ elementwise such that $|V(s) - \mathcal{T}V(s)| = \mathcal{T}V(s) - V(s)$. Similarly,

$$\begin{aligned}\left|V^\dagger(s) - \mathcal{T}_{\pi^\dagger}V(s)\right| &= |V(s) - \mathcal{T}_{\pi^\dagger}V(s)| \\ &= \mathcal{T}_{\pi^\dagger}V(s) - q(s) + q(s) - V(s) \\ &\leq 2\gamma\epsilon + 2\gamma\epsilon \\ &= 4\gamma\epsilon.\end{aligned}$$

Now, we prove in either case we have $||V^\dagger - \mathcal{T}V||_\infty \leq 4\gamma\epsilon$ and $||V^\dagger - \mathcal{T}_{\pi^\dagger}V||_\infty \leq 4\gamma\epsilon$. □

Remark. Since the quality of the Monotonic Random Value Operator does not change too much. We can use the idea of Sublinear Randomized Value Iteration to get a better policy.

Algorithm 8 Monotonic Sampled Randomized Value Iteration: *SampledRandomizedMonVI*$(V_0, \pi_0, L, \epsilon, \delta)$

Input: initial value V_0, initial policy $\pi \in \mathcal{A}^{|\mathcal{S}|}$ with $V_0 \leq \mathcal{T}_{\pi_0}V_0$, and number of iterations $L > 0$
Input: target accuracy $\epsilon > 0$ and failure probability $\delta \in (0, 1)$

1: Sample n samples to obtain approximate offsets: $\mathcal{X} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}$ with $|\mathcal{X}(s,a) - \mathbb{E}_{s'\sim p(\cdot|s,a)}[V_0(s')]| \leq \epsilon$ for all (s, a).

$$n = \left\lceil \frac{2||V_0||_\infty^2}{\epsilon^2} \log\left(\frac{2L}{\delta}\right)\right\rceil.$$

2: **for** each round $\ell = 1, 2, \cdots L$ **do**
3: $\quad (V_\ell, \pi_\ell) = ApxMonVal(V_{\ell-1}, \pi_{\ell-1}, V_0, \mathcal{X}, \epsilon/2, \delta/L)$.
4: **end for**

Output: (V_L, π_L).

Algorithm 9 Monotonic Sublinear Randomized Value Iteration: *SublinearRandomizedMonVI*(ϵ, δ)

Input: desired precision ϵ and failure probability $\delta \in (0, 1)$.

1: Set $K = \left\lceil \log_2\left(\frac{1}{\epsilon(1-\gamma)}\right)\right\rceil$, and $L = \left\lceil \frac{1}{1-\gamma}\log\left(\frac{4}{1-\gamma}\right)\right\rceil$
2: Set $V_0 = \vec{0}$, $\pi_0 \in \mathcal{A}^{|\mathcal{S}|}$ arbitrary, and $\epsilon_0 = \frac{1}{1-\gamma}$.
3: **for** each iteration $k = 1, 2, \cdots, K$ **do**
4: $\quad$ Set $\epsilon_k = \frac{1}{2}\epsilon_{k-1} = \frac{1}{2^k(1-\gamma)}$.
5: $\quad (V_k, \pi_k) = SampledRandomizedMonVI(V_{k-1}, L, (1-\gamma)\epsilon_k/(4\gamma), \delta/K)$.
6: **end for**

Output: (V_K, π_K).

Theorem 3.11 (Quality of Monotonic Sublinear Randomized Value Iteration)**.** *With probability at least $1-\delta$, the output policy/(value function) of Monotonic Sublinear Randomized Value Iteration (see Algorithm 9) is ϵ-optimal with the following number of samples*

$$\tilde{\mathcal{O}}\left(\frac{|\mathcal{S}| \times |\mathcal{A}|}{\epsilon^2(1-\gamma)^4}\log\left(\frac{1}{\delta}\right)\right).$$

Proof. The basic proof procedure is almost same with the one of Theorem 3.4 except that we need to 1) ensure all assumed conditions holds; 2) approximately adjust parameters (that has been highlighted with red in the corresponding algorithm procedures). First, we note that the conditions of *ApxMonVal* hold recursively by Lemma 3.9 and this holds trivially at the initialization of Algorithm 9. In addition, since V_ℓ is improved monotonically, we have

$$\|V_\ell - V^*\|_\infty \leq \|V_0 - V^*\|_\infty, \text{ for all } \ell = 1, \cdots, L.$$

In this way, we do not use the estimate range in Equation (3.1) and correspondingly the bound does not include the additional $(1-\gamma)^{-3}$ term. □

Remark. The benefits of monotonic operator are summarized below. For the quality of value function, compared to Theorem 3.4 in which we do not know $\epsilon^{-2}(1-\gamma)^{-4}$ or $(1-\gamma)^{-3}$ dominates in practice, we guarantee that only $\epsilon^{-2}(1-\gamma)^{-4}$ dominates. For the quality of the induced policy, we guarantee that the sample complexity is improved from $\mathcal{O}((1-\gamma)^{-6})^4$ to $\mathcal{O}((1-\gamma)^{-4})$.

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[4]This can obtained from Theorem 3.4 by substituting ϵ with $(1-\gamma)^{-1}\epsilon$.

Appendix A Proofs in Section 2

A.1 Proof of Lemma 2.7

Proof. For the first inequality,

$$\sigma^2_{V^*}(s,a) = \underbrace{\left\{\gamma^2 \mathbb{V}_{s'\sim p(\cdot|s,a)}[V^*(s')] - \gamma^2 \mathbb{V}_{s'\sim \hat{p}(s,a)}[V^*(s')]\right\}}_{(a)} + \underbrace{\left\{\gamma^2 \mathbb{V}_{s'\sim \hat{p}(\cdot|s,a)}[V^*(s')]\right\}}_{(b)}.$$

For the first term (a), we show that for any (s,a),

$$\begin{aligned}
&\gamma^2 \mathbb{V}_{s'\sim p(\cdot|s,a)}[V^*(s')] - \gamma^2 \mathbb{V}_{s'\sim \hat{p}(\cdot|s,a)}[V^*(s')] \\
&= \gamma^2 \left[P^{\pi^*} - \hat{P}^{\pi^*}\right] V^{*2}(s,a) - \gamma^2 \left[\left(P^{\pi^*} V^*\right)^2 (s,a) - \left(\hat{P}^{\pi^*} V^*\right)^2 (s,a)\right] \\
&= \gamma^2 \underbrace{\left[P^{\pi^*} - \hat{P}^{\pi^*}\right] V^{*2}(s,a)}_{(i)} - \gamma^2 \underbrace{\left[\left(P^{\pi^*} - \hat{P}^{\pi^*}\right) V^*\right]}_{(ii)} \underbrace{\left[\left(P^{\pi^*} + \hat{P}^{\pi^*}\right) V^*\right]}_{(iii)}.
\end{aligned}$$

We can see that both (i) and (ii) are contractive zero-mean random variables, and (iii) can be upper bounded since $||P^{\pi^*}||_\infty$ and $||P^{\pi^*}||_\infty$ are bounded by 1. More precisely, applying the Hoeffding's inequality on the terms (i) and (ii), we have with probability $1-\delta$, the following inequalities hold separately:

$$\begin{aligned}
\left[P^{\pi^*} - \hat{P}^{\pi^*}\right] V^{*2}(s,a) &\leq \frac{1}{(1-\gamma)^2}\sqrt{\frac{2\log\frac{1}{\delta}}{n}}, \\
\left[P^{\pi^*} - \hat{P}^{\pi^*}\right] V^{*}(s,a) &\geq \frac{1}{1-\gamma}\sqrt{\frac{2\log\frac{1}{\delta}}{n}}.
\end{aligned}$$

Note that

$$\left[\left(P^{\pi^*} + \hat{P}^{\pi^*}\right) V^*\right] \leq \frac{2}{1-\gamma}.$$

Thus, we conclude that with probability at least $1-\delta$, for any (s,a),

$$\gamma^2 \mathbb{V}_{s'\sim p(\cdot|s,a)}[V^*(s')] - \gamma^2 \mathbb{V}_{s'\sim \hat{p}(s,a)}[V^*(s')] \leq \frac{3\gamma^2}{(1-\gamma)^2}\sqrt{\frac{2\log\frac{2}{\delta}}{n}}.$$

For the second term (b), we have (for simplicity, we omit the subscript $s' \sim \hat{p}(s,a)$ when writing the

expectation): for any (s,a)

$$
\begin{aligned}
\sigma^2_{V^*}(s,a) &= \gamma^2 \mathbb{E}\left[\{V^* - \mathbb{E}[V^*]\}^2\right] \\
&= \gamma^2 \mathbb{E}\Bigg[\Bigg\{ \underbrace{\left(V^* - \hat{V}^{\pi^*}\right) - \left(\mathbb{E}[V^*] - \mathbb{E}[\hat{V}^{\pi^*}]\right)}_{V^\dagger} + \underbrace{\left(\hat{V}^{\pi^*} - \mathbb{E}[\hat{V}^{\pi^*}]\right)}_{V_o} \Bigg\}^2\Bigg] \\
&= \gamma^2 \left(\mathbb{E}\left[V_\dagger^2\right] + \mathbb{E}\left[V_o^2\right] + 2\mathbb{E}\left[V_\dagger V_o\right]\right) \\
&\overset{(1)}{\leq} \gamma^2 \left(\mathbb{E}\left[V_\dagger^2\right] + \mathbb{E}\left[V_o^2\right] + 2\sqrt{\mathbb{E}\left[V_\dagger^2\right]\mathbb{E}\left[V_o^2\right]}\right) \\
&= \gamma^2 \left(\sqrt{\mathbb{E}\left[V_\dagger^2\right]} + \sqrt{\mathbb{E}\left[V_o^2\right]}\right)^2 \\
&= \gamma^2 \left\{\sqrt{\mathbb{V}_{s'\sim\hat{p}(s,a)}\left[V^*(s') - \hat{V}^{\pi^*}(s')\right](s,a)} + \sqrt{\mathbb{V}_{s'\sim p(\cdot|s,a)}\left[\hat{V}^{\pi^*}(s')\right](s,a)}\right\}^2 \\
&\overset{(2)}{\leq} \left\{||V^* - \hat{V}^{\pi^*}||_\infty + \hat{\sigma}^2_{\hat{V}^{\pi^*}}(s,a)\right\}^2.
\end{aligned}
\tag{A.1}
$$

where step (1) follows the Cauchy-Schwarz inequality for the expectation that $|\mathbb{E}[(XY)]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$ for two random variables X and Y, and step (2) follows that for any (s,a),

$$
\mathbb{V}_{s'\sim\hat{p}(s,a)}\left[V^*(s') - \hat{V}^{\pi^*}(s')\right](s,a) \leq ||V^* - \hat{V}^{\pi^*}||^2_\infty.
$$

Clearly, the bound of $||V^* - \hat{V}^{\pi^*}||_\infty$ can be derived by the Hoeffding's inequality (ref to the proof of Theorem 2.4). That is, with probability at least $1-\delta$, we have

$$
||V^* - \hat{V}^{\pi^*}||_\infty \leq \sqrt{\frac{4\gamma^2 \log\left(\frac{2|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^4}}.
$$

Return back to Equation (A.1), we have with probability at least $1-\delta$, for all (s,a)

$$
\sigma^2_{V^*}(s,a) \leq \left[\hat{\sigma}_{\hat{V}^{\pi^*}}(s,a) + \sqrt{\frac{4\gamma^2 \log\left(\frac{6|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n(1-\gamma)^4}}\right]^2 + \frac{3\gamma^2}{(1-\gamma)^2}\sqrt{\frac{2\log\left(\frac{3|\mathcal{S}|\times|\mathcal{A}|}{\delta}\right)}{n}}.
$$

Moving the last term to the RHS and taking the root yields the desired result. □

A.2 Proof of Lemma 2.10

Proof. The first inequality follows Lemma 2.9 by solving the Bellman-like equation and the upper bound $||\Sigma^\pi||$ follows its definition (see Equation (2.2.2)). For the second inequality, we have

$$\begin{aligned}
\left\|(\mathbb{I}-\gamma P^\pi)^{-1}\,\sigma_{V^\pi}\right\|_\infty &= \left\|\sum_{k\geq 0}(\gamma P^\pi)^k\sigma_{V^\pi}\right\|_\infty \\
&= \left\|\sum_{\ell\geq 0}(\gamma P^\pi)^{t\ell}\sum_{j=0}^{t-1}(\gamma P^\pi)^j\sigma_{V^\pi}\right\|_\infty \\
&\leq \sum_{\ell\geq 0}(\gamma^t)^\ell\left\|\sum_{j=0}^{t-1}(\gamma P^\pi)^j\sigma_{V^\pi}\right\|_\infty \\
&= \frac{1}{1-\gamma^t}\left\|\sum_{j=0}^{t-1}(\gamma P^\pi)^j\sigma_{V^\pi}\right\|_\infty,
\end{aligned}$$

in which we write $k = t\ell + j$ for any positive integer t. Then we have

$$\begin{aligned}
\left\|\sum_{j=0}^{t-1}(\gamma P^\pi)^j\sigma_{V^\pi}\right\|_\infty &\overset{(i)}{\leq} \left\|\sum_{j=0}^{t-1}\gamma^j\sqrt{(P^\pi)^j\sigma^2_{V^\pi}}\right\|_\infty \\
&\overset{(ii)}{\leq} \sqrt{t}\left\|\sqrt{\sum_{j=0}^{t-1}\left(\gamma^2P^\pi\right)^j\sigma^2_{V^\pi}}\right\|_\infty \\
&\leq \sqrt{t}\left\|\sqrt{\left(\mathbb{I}-\gamma^2P^\pi\right)^{-1}\sigma^2_{V^\pi}}\right\| \\
&= \left\|\sqrt{t\Sigma^\pi}\right\|,
\end{aligned}$$

where step (i) uses Jensen's inequality that $\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}$ for a random variable X, and step (ii) use the Cauchy-Scwarz inequality that $\sum_{i=1}^t \sqrt{x_i} \leq \sqrt{t\sum_{i=1}^t x_i}$. Based on the estimate of Σ^π, we have that

$$\left\|\left(\mathbb{I}-\gamma P^\pi\right)^{-1}\sigma_{V^\pi}\right\|_\infty \leq \frac{\sqrt{t}}{(1-\gamma^t)}\sqrt{\Sigma^\pi}.$$

Since the above inequality holds for every positive t, we can choose $t = \text{int}(\frac{1}{1-\gamma})$, and note that $\frac{1}{1-\gamma^{1/(1-\gamma)}}$ is increasing with respect to γ and can be bounded by 2 (you can convince by plotting the curve of this function, see Figure A.2). In this way, we get the desired result.

□

A.3 Proof of Theorem 2.11

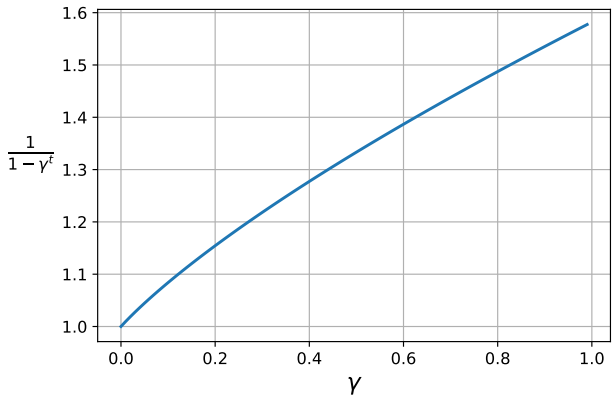

Figure 1: Curve of $1/(1-\gamma^t)$ when $t = \frac{1}{1-\gamma}$.

Appendix B Proofs in Section 3

B.1 Proof of Lemma 3.6

Proof. In Section 1.3, we have proved that Bellman (Optimality) Operator $\mathcal{T}$ is γ-contractive. Here we prove that $\mathcal{T}_\pi$ is also γ-contractive, which will be used later.

Lemma B.1 (γ-contractive of $\mathcal{T}_\pi$)**.** *For any deterministic policy $\pi \in \mathcal{A}^{|\mathcal{S}|}$, define the operator $\mathcal{T}_\pi$:*

$$\mathcal{T}_\pi(V)(s) = r(s, \pi(a)) + \gamma \mathbb{E}_{s' \sim p(\cdot|s,\pi(s))}[V(s')].$$

We have that $\mathcal{T}_\pi$ is γ-contractive, that is, for any vector $V_1, V_2 \in \mathbb{R}^{|\mathcal{S}|}$,

$$\|\mathcal{T}_\pi(V_1) - \mathcal{T}_\pi(V_2)\|_\infty \leq \gamma \|V_1 - V_2\|_\infty .$$

Proof.

$$\begin{aligned}
&\|\mathcal{T}_\pi(V_1) - \mathcal{T}_\pi(V_2)\|_\infty \\
&\leq \max_{s\in\mathcal{S}} \left| \left(r(s,\pi(a)) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V_1(s')]\right) - \left(r(s,\pi(a)) + \gamma \mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V_2(s')]\right) \right| \\
&\leq \gamma \max_{s\in\mathcal{S}} \left| \mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V_1(s')] - \mathbb{E}_{s'\sim p(\cdot|s,\pi(s))}[V_2(s')] \right| \\
&\leq \gamma \|V_1 - V_2\|_\infty .
\end{aligned}$$

□

As a consequence of Lemma B.1, V_π is the fixed point of $\mathcal{T}_\pi$, that is,

$$V_\pi = \mathcal{T}_\pi V_\pi.$$

Back to Lemma 3.6, we know that

$$\begin{aligned}
\|V_\pi - \mathcal{T}_\pi V\|_\infty &= \|\mathcal{T}_\pi V_\pi - \mathcal{T}_\pi(\mathcal{T}_\pi V) + \mathcal{T}_\pi(\mathcal{T}_\pi V) - \mathcal{T}_\pi V\|_\infty \\
&\leq \|\mathcal{T}_\pi V_\pi - \mathcal{T}_\pi(\mathcal{T}_\pi V)\|_\infty + \|\mathcal{T}_\pi(\mathcal{T}_\pi V) - \mathcal{T}_\pi V\|_\infty \\
&\leq \gamma \|V_\pi - \mathcal{T}_\pi V\|_\infty + \gamma \|\mathcal{T}_\pi V - V\|_\infty .
\end{aligned}$$

By rearranging, we get the desired result

$$\|V_\pi - \mathcal{T}_\pi V\|_\infty \leq \frac{\gamma}{1-\gamma} \|\mathcal{T}_\pi V - V\|_\infty .$$

□